7 Grey gurnard in the North Sea, eastern English Channel, Skagerrak, and Kattegat

gug.27.3a47d – *Eutrigla gurnardus* in Subarea 4 and divisions 7.d and 3.a

7.1 General

Grey gurnard (*Eutrigla gurnardus*) was assessed in the Working Group on the Assessment of New MoU Species (ICES, 2014) until 2014. Since 2015, the stock was assessed by the Working Group on the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK), and defined as a category DLS 3.2 stock (ICES, 2015). For this stock, only survey data and limited catch data (InterCatch data 2012–2022) were available. Official landings data are incomplete or were not reported specifically for grey gurnard in the past. Grey gurnard in Subarea 4, Divisions 7.d and 3.a is a non-target stock with no TAC, and has a biennial advice. ICES has not been requested to provide advice on fishing opportunities for this stock for the years 2019–2022. However, a new catch advice was requested for the years 2023 and 2024, and during WGNSSK 2022, the empirical *rb* rule (ICES, 2020b) was applied for the first time as the basis for the catch advice. During WGNSSK 2024, the assessment was updated and the *rb* rule was applied again as basis for the catch advice 2025-2026. The advice was 4092 tonnes, 30% less than the previous catch advice (2023-2024). During the WGNSSK 2025 no new catch advice was due and only catch and survey data were updated.

7.1.1 Biology and ecosystem aspects

Grey gurnard (*Eutrigla gurnardus*) occurs in the eastern Atlantic, from Iceland, Norway, the southern Baltic, and North Sea to southern Morocco and Madeira. It is also found in the Mediterranean and Black Seas. In the North Sea and in the Skagerrak/Kattegat, grey gurnard is an abundant demersal species. In the North Sea, the species may form dense semi-pelagic aggregations in winter to the northwest of the Dogger Bank, whereas in summer it is more widely distributed. The species is less abundant in the Channel, the Celtic Sea and in the Bay of Biscay. Spawning takes place in spring and summer. Clear nursery areas have not been identified.

Grey gurnard is considered a predator on young age groups of a number of commercially important demersal stocks (cod, whiting, haddock, sandeel, and Norway pout) in the North Sea (de Gee and Kikkert, 1993). Therefore, an increase in abundance of grey gurnard can lead to an increase in mortality especially of North Sea cod (age-0) and whiting (age-0 and age-1) (ICES, 2017). The multispecies model SMS estimated that grey gurnard can cause up to 50% of the predation mortality on 0-group cod and whiting. Therefore, the abundance and distribution pattern of grey gurnard and its prey size preferences are highly relevant from an ecological point of view (Floeter and Temming, 2005; Kempf *et al.*, 2013).

7.1.2 Stock ID and possible assessment areas

No studies are known on the stock ID of grey gurnard. In a pragmatic approach for advisory purposes and in order to facilitate addressing ecosystem considerations, the population is currently split among three ecoregions: North Sea including Division 7.d, Celtic Seas and the South

European Atlantic. This proposal should be discussed considering the low levels of catches reported in recent years in Celtic Seas and the South European Atlantic (ICES, 2011; ICES, 2012).

7.1.3 Management regulations

There is no minimum landing size for this species and there is no TAC.

7.2 Fisheries data

7.2.1 Historical landings

Historically, grey gurnard is taken as a by-catch species in mixed demersal fisheries for flatfish and round fish. Grey gurnard from the North Sea is mainly landed for human consumption purposes. However, the market is limited and the largest part of the catch is discarded. Owing to the low commercial value of this species and the high observed discard rates, official reported landings data do not reflect the actual catches.

In the past, gurnards were often not sorted by species when landed and were reported as one generic category of “gurnards”. Furthermore, catch statistics are incomplete for some years, e.g. the Netherlands did not report gurnards during the years 1984–1999. In recent years, the official statistics seem to improve gradually. However, some countries continue to report “gurnards” landings and do not provide information on grey gurnard separately for all cases, or the data imported into InterCatch are based on a gurnard mix, raised by survey information on the proportion of the specific gurnard species. The latter approach is highly questionable, because tub gurnard is a much more valuable bycatch species and a larger proportion of that species is landed compared to grey gurnard.

Since the early 1980s, specific landings data for grey gurnard are available from the official catch statistics. Before that, these data occurred only sporadically in the statistics. Most of grey gurnard catches are taken in Subarea 4 and to a much lesser extent in divisions 7.d and 3.a (Figure 7.1–7.3; Table 7.3–7.7). Exceptionally high annual landings were reported during the late 1980s to early 1990s with a maximum of 46 598 tonnes in 1987 (Figure 7.2; Table 7.6) because of Danish landings for reduction purposes. After this peak, the Danish landings dropped again to low levels. Compared to 2023 (869 tonnes), the total official landings in 2024 with 711 tonnes were slightly lower but on a similar level, and were again among the lowest observed for the recent ten years. The average official landings for the last ten years (2015–2024) was 1556 tonnes. Official landings data from 1950 to 2005 were taken from the “ICES catch statistics 1950 to 2010” (https://www.ices.dk/data/Documents/CatchStats/HistoricalLandings1950-2010.zip). Data from 2006 to 2022 were taken from the “ICES catch statistics 2006 to 2022” (https://www.ices.dk/data/Documents/CatchStats/OfficialNominalCatches.zip). Data for 2023 and 2024 were taken from the preliminary catch statistics provided by ICES (http://data.ices.dk/rec12/login.aspx).

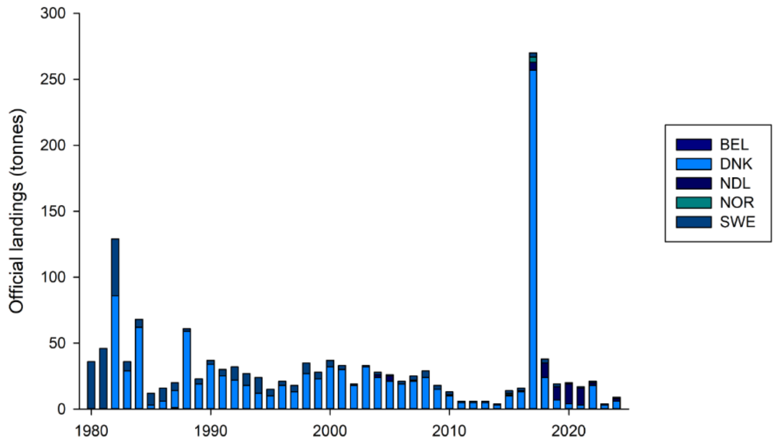

Figure 7.1. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Official landings of grey gurnard in Division 3.a 1980–2024.

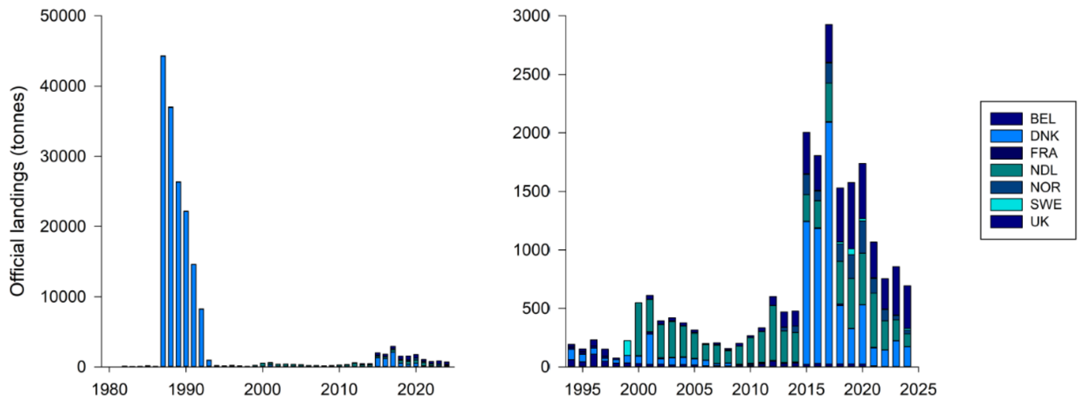

Figure 7.2. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Official landings of grey gurnard in Subarea 4 by country for the years 1980 - 2024 (a), and official landings of grey gurnard by country in Subarea 4 for the years 1994 - 2024 (b).

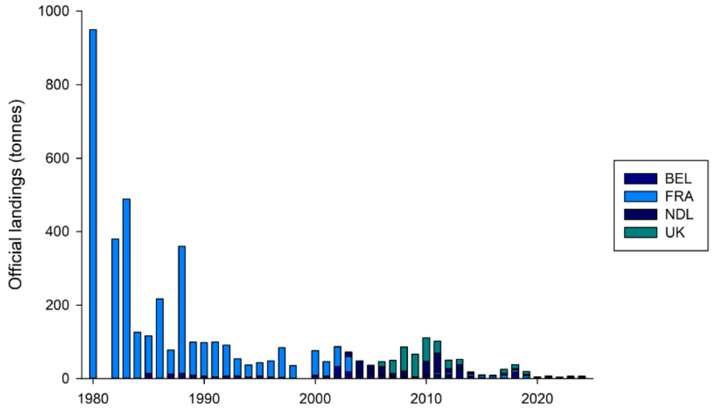

Figure 7.3. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Official landings by country of grey gurnard in Division 7.d for the years 1980–2024.

7.2.2 InterCatch data

InterCatch contains data for the years 2012–2024. UK England data updates for the years 2021-2023 were taken into account and only changed the total catch slightly (-8 tonnes in 2021; +291 tonnes in 2022; +98 tonnes in 2023). The largest amount of landings in 2024 was reported by Scotland (327 tonnes, OTB_DEF_>=120_0_0_all; Figure 7.4). Considerable amounts of landings were also reported by Denmark as industrial bycatch (MIS_MIS_0_0_0_IBC, 171 tonnes), and by several Dutch fleets (e.g., SSC_DEF_70-99_0_0_all, 49 tonnes). For all countries, the amount of discards exceeded the amount of landings (Figure 7.7 and 7.8). The largest amounts of discards were reported for the Scottish demersal otter trawl fleet (OTB_DEF_>=120_0_0_all, 569 tonnes), the Dutch TBB_DEF_70-99_0_0_all fleet (424 tonnes), Dutch OTB_CRU_70-99_0_0_all fleet (81 tonnes), and the Danish SSC_DEF_>=120_0_0_all fleet (52 tonnes) (Figure 7.6).

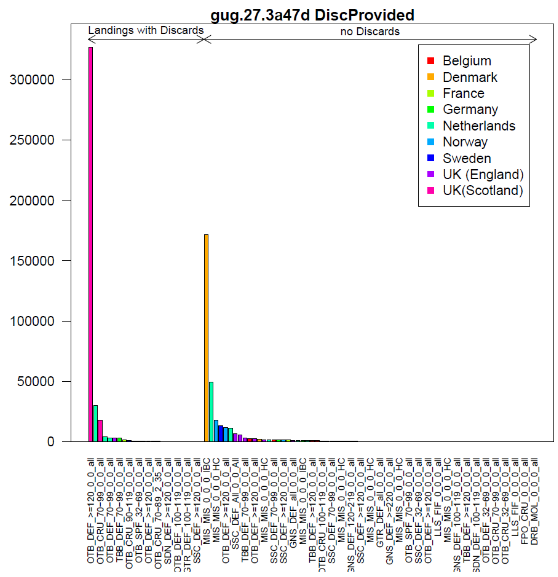

Figure 7.4. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Grey gurnard landings (kg) in 2024 by fleet and country as uploaded to InterCatch, with indication of discard provision.

In total, 56% of landings were reported with discards (Figure 7.5). For all other métiers with landings, discards were raised within InterCatch. A number of métiers reported zero landings for which no discard raisings were possible. However, most of these métiers are probably negligible (e.g. FPO_CRU_0_0_0_all). But it has to be noted, that grey gurnard is probably also a considerable part of unwanted bycatch in some of these "zero landing métiers".

For the discard raising, it was attempted to use the same groupings as for the previous data years. However, this was not possible for all cases and compared to the previous year, slight changes had to be made. The grouping is based on gear type and mesh size over areas and season. For the length sampling allocation, only one landings-group and one discard-group was set up, because data availability did not allow for a higher resolution. Length samples for landings were provided only by one métiers for 2024 data (SCO OTB_DEF_>=120_0_0_all), and this was allocated subsequently for all other landing métiers.

The following groupings were used for the 2024 data discard raising:

Group 1: all passive gears -> raised with GTR_DEF_all_0_0_all SWE. No other métier available for passive gear.

Group 2: MIS_MIS_0_0_0_HC -> no discard data available for this métier. Raised with all other métiers.

Group 3: all TBB_DEF_70-99 métiers -> raised with all TBB_DEF_70-99 métiers

Group 4: OTB_CRU_70-99_0_0_all -> raised with all OTB_CRU_70-99_0_0_all

Group 5: OTB_DEF_120_0_0_all -> raised with all OTB_DEF_120_0_0_all

Group 6: OTB_DEF_100-119_0_0_all, SSC_DEF_100-119_0_0_all, TBB_DEF_100-119_0_0_all -> raised with all other métiers with 100-119.

Group 7: OTB_DEF_70-99_0_0, SSC_DEF_70-99_0_0_all, SDN_DEF_70-99_0_0_all, OTM_SPF and OTB_SPF_70-99_0_0_all -> raised with all OTB_DEF_70-99_0_0_all

Group 8: SSC_DEF_>=120_0_0_all and SDN_DEF_>=120_0_0_all -> raised with SSC_DEF_>=120_0_0_all and SDN_DEF_>=120_0_0_all

Group 9: SSC_DEF_32-69_0_0_all, OTB_DEF_32-69_0_0_all, OTB_CEP_32-69_0_0_all, OTB_SPF_32-69_0_0_all, and PS_SPF_0_0_0 (FRA) -> raised with all available métiers with 32-69_0_0_all.

Some métiers were not raised because no suitable data were available or they were negligible:

- MIS_MIS_0_0_0_IBC (14 métiers)
- OTB_CRU_16-31_0_0_all (6 métiers)

The largest amount of discards were raised for the Dutch SSC_DEF_70-99_0_0_all fleet (589 tonnes), the UK(Eng) SSC_DEF_All_0_0_All fleet (76 tonnes), the UK(Eng) OTB_DEF_70-99_0_0_all fleet (66 tonnes). The total catch estimated with InterCatch for the year 2024 was 3261 tonnes from which 707 tonnes were landings and 2554 tonnes estimated discards. The Netherlands took the largest proportion of the total catch in 2024, followed by UK England, and UK Scotland (Figure 7.8). An overview of ICES catches, discard rate and official landings is shown in Table 7.9.

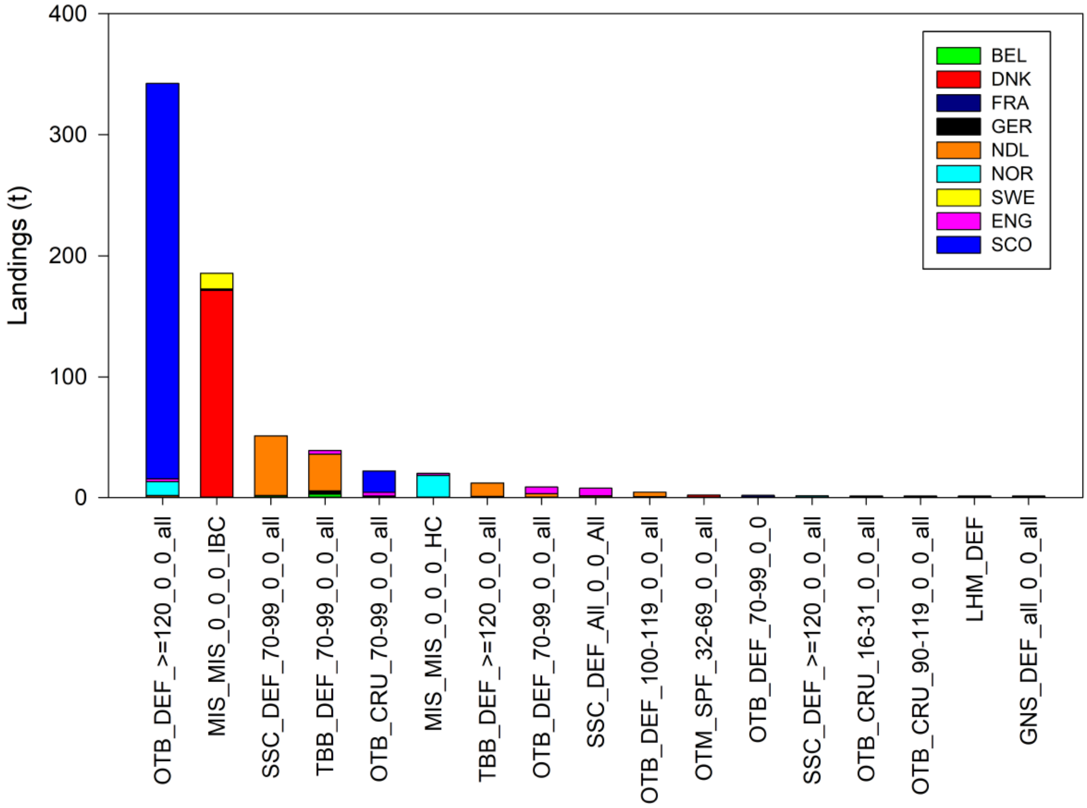

Figure 7.5. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. InterCatch landings by fleet and country (2024 data).

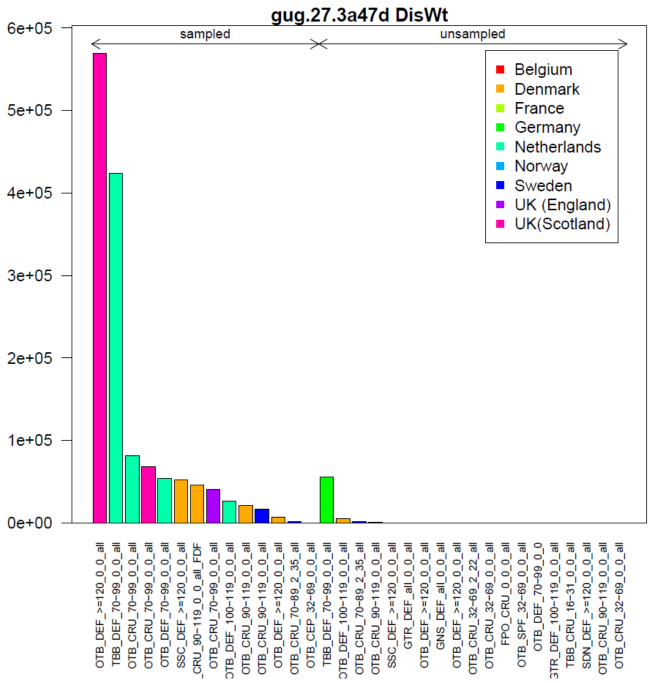

Figure 7.6. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Grey gurnard discards (kg) in 2024 by fleet and country as uploaded to InterCatch, with indication of sampled strata for length.

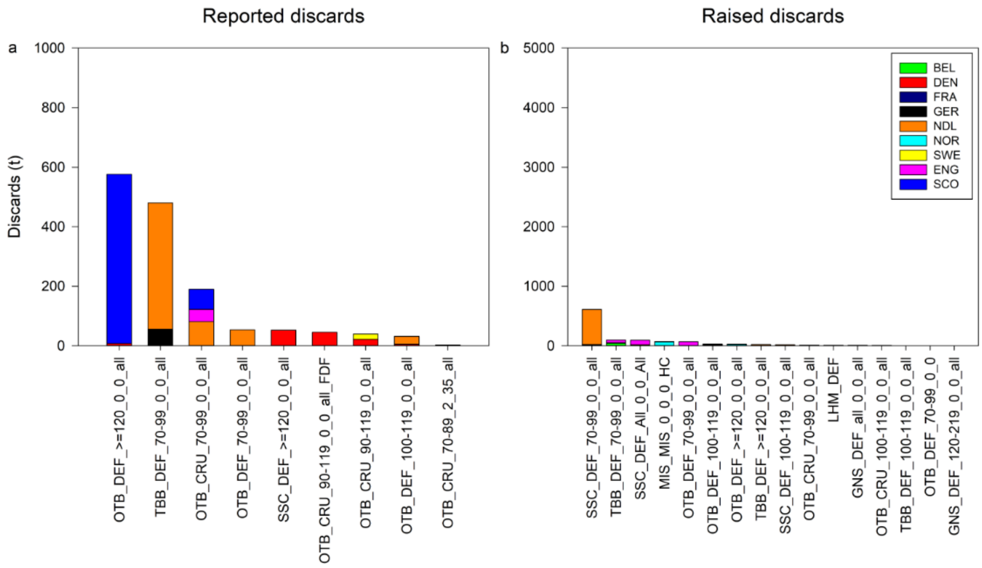

Figure 7.7. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Grey gurnard discards in 2024 by fleet and country. Reported discards panel (a), raised discards panel (b). Legend valid for both panels.

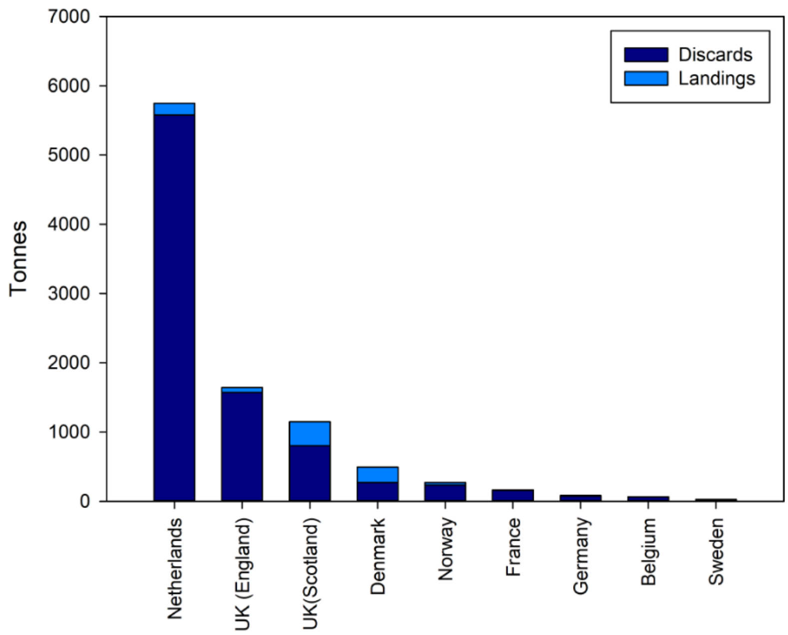

Figure 7.8. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Grey gurnard total ICES estimated landings and discards in 2024 by country.

7.2.3 Other information on Discards

In Table 7.1, the numbers per hour of discarded grey gurnard in Dutch bottom-trawl fisheries in the North Sea and eastern Channel are shown for 2006–2012 (Uhlmann *et al.*, 2013). The rates are highly variable depending on the specific métiers, with highest values observed for the SSC_DEF métiers. German discard data from an observer programme indicate that the proportion of discarded gurnard in German demersal trawl fisheries ranges between 76.6% and 93.0% (Ulleweit *et al.*, 2010).

Table 7.1 Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Discards in numbers per hour of grey gurnard by different métiers in the Netherlands 2006–2012.

Métier	TBB_DEF	TBB_DEF*	TBB_DEF	SSC_DEF	SSC_DEF	OTB_MCD	OTB_DEF	OTB_DEF
Mesh	70-99	70-99	100-119	100-119	>120	70-99	70-99	100-119
2006	68.3							
2007	60.2							
2008	34.3							
2009	55	17	37			111	77	15
2010	81	10	109			47	52	110
2011	61	27	10	NA	119	27	55	70
2012	41	24	30	317	307	110	75	12

*≤300 hp segment

7.3 Survey data/recruit series

For the North Sea and Skagerrak/Kattegat, data are available from the International Bottom Trawl Survey in the North Sea (NS-IBTS; ICES, 2021a). The NS-IBTS Q1 and NS-IBTS Q3 can provide information on distribution and the length composition of the stock. Grey gurnard occurs throughout the North Sea and Skagerrak/Kattegat. During winter, grey gurnards are concentrated to the northwest of the Dogger Bank at depths of 50–100 m, while densities are lower off the Danish coast, in the German Bight and eastern part of the Southern Bight (Figure 7.9). The distribution pattern changes substantially in spring, when the whole area south of 56°N becomes more densely populated and the high concentrations in the central North Sea disappear until the next winter (Daan *et al*., 1990; Figure 7.10).

The near absence of grey gurnard in the southern North Sea during winter and the marked shift in the centre of distribution between winter and summer suggests a preference for higher water temperatures (Hertling, 1924; Daan *et al*., 1990).

During winter, grey gurnard occasionally form dense aggregations just above the seabed (or even in midwater, especially during nighttime) which may result in extremely large catches. Within one survey, these large hauls may account for 70% or more of the total catch of all species. Bottom temperatures in high density areas usually range from 8 to 13°C (Sahrhage, 1964).

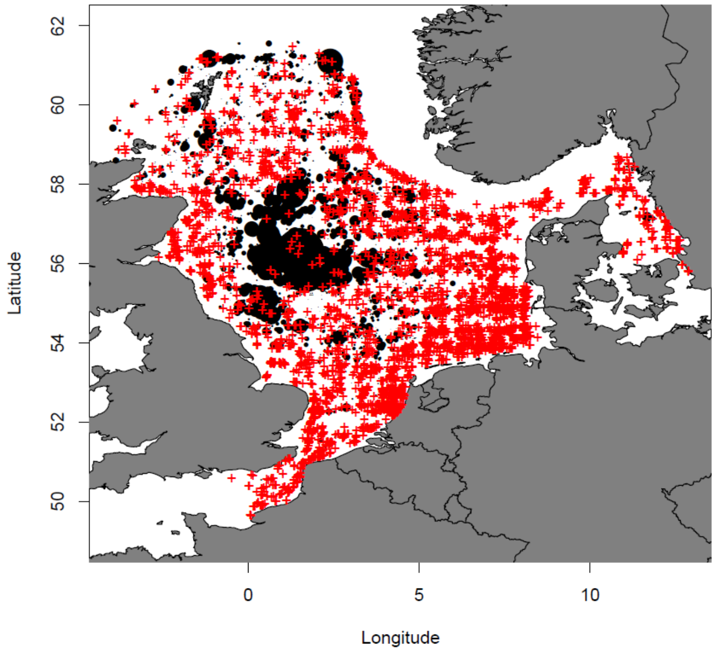

Figure 7.9. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Spatial distribution of grey gurnard from the NS-IBTS Q1 survey (all years) in Subarea 4 and Division 3.a. Red crosses display hauls with no catches of grey gurnard.

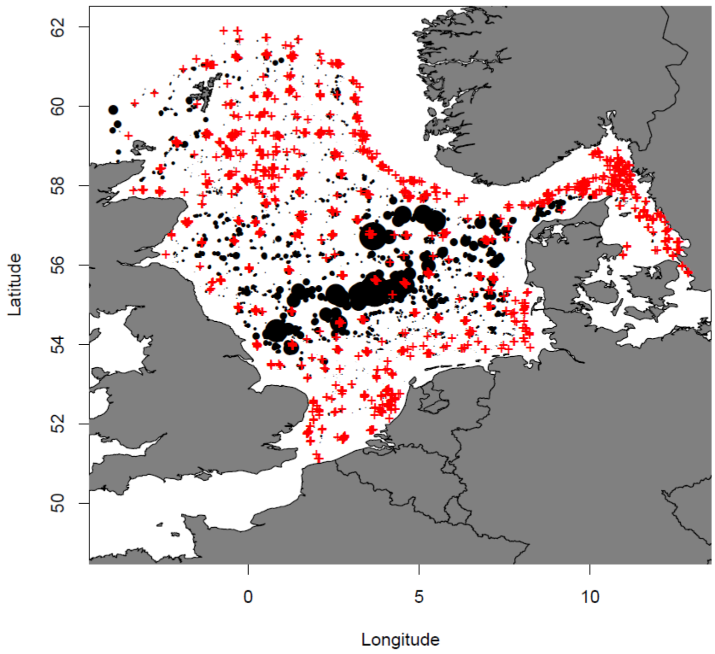

Figure 7.10. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Spatial distribution of grey gurnard from the NS-IBTS Q3 survey (all years) in Subarea 4 and Division 3.a. Red crosses display hauls with no catches of grey gurnard.

7.4 Biological sampling

Individual biological data for this species are scarce (see also stock annex). In the North Sea, individual biological data have been collected sporadically during some years of the NS-IBTS Q1 and NS-IBTS Q3 survey. The age readings done on collected otoliths from NS-IBTS Q1 resulted in an age range from 2 to 14 years, but not that many individuals were aged (n = 469, years 2010 and 2014).

Available data on grey gurnard individual weights and maturity were analysed in order to estimate a mature biomass index. The obtained weight–length relation was:

Weight = 0.006 * LngtClass ^ 3.082 (NS-IBTS Q1 and Q3 2010-2014; 2016-2018 data; Figure 7.11a).

A maturity ogive based on all available grey gurnard maturity data from NS-IBTS Q1 was used to calculate this mature biomass index. The obtained maturity ogive shows that above 21.1 cm more than 95% of all the individuals can be considered mature (Figure 7.11b; DATRAS CA maturity data were available for the years 1992, 2010, 2014, 2016). The corresponding Lmat50% value was 16.3 cm. Proportion mature at length was calculated by the model:

Prop-Mat = 0.991 / (1 + exp (-1 * (LngtClass - 16.273) / 2.105)).

The available age and maturity data suggest that grey gurnard is maturing early in the North Sea and a certain proportion of fish at age 1 are mature.

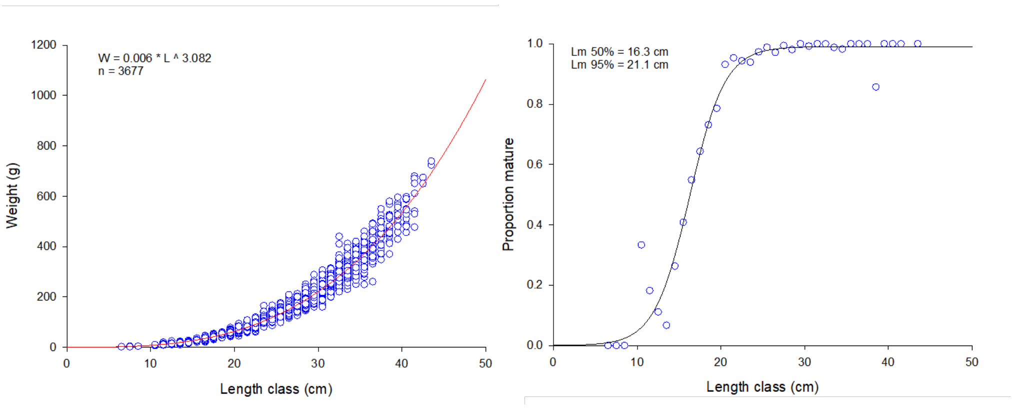

Figure 7.11 Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Length-weight relationship from NS-IBTS Q1 and IBTS Q3 CA data (left panel); maturity ogive obtained from NS-IBTS Q1 CA data (right panel).

7.5 Analysis of stock trends/assessment

Information from landings is poor, due to poor reporting (gurnard species were not always identified in the data, and probably also misreporting has occurred), and because the low value of this species leads to massive discarding.

To analyse stock trends, a mature biomass index was calculated applying a length-weight relationship and a maturity ogive which were obtained from available NS-IBTS CA (DATRAS) records (see Section 7.4 above).

According to van Heesen and Daan (1996), outliers were excluded from previous NS-IBTS Q1 time-series, since grey gurnards tend to form dense concentrations during winter. Outliers were defined as hauls which accounted for more than 90% of the total gurnard weight caught in the respective year. However, such extreme outliers were only identified in the time period before 1983 which is not displayed here. The start of the time-series chosen here is 1983, since full standardization of the NS-IBTS was reached then. The time-series of mature biomass index of grey gurnard of the NS-IBTS Q1 survey has shown a strong increasing trend from the beginning of 1990s (Figure 7.12; Table 7.8). Since then, it was fluctuating on a high level until 2017. A strong decline of the index was observed for the year 2018. In 2019, the index value was only slightly higher. The NS-IBTS Q1 index then decreased further until 2022. The 2023 value increased again, but still on comparable low level, while the values for 2024 and 2025 were again somewhat lower and nearly the same. The mature biomass index for the NS-IBTS Q3 does not show the same pronounced increasing trend compared to the quarter 1 index. The 2014 value was the highest observed in this time-series. Since then, the NS-IBTS Q3 index showed an overall decreasing trend until 2024. In general, lower biomass and abundance values were observed for the NS-IBTS Q3 survey time-series. Compared to the North Sea/Skagerrak (Subarea 4/Division 3.a), the mature biomass values recorded by the Channel Ground Fish Survey (CGFS) in the eastern English Channel (Division 7.d) were extremely low (not shown in this report). No trend could be detected in the CGFS index. Therefore, the advice for grey gurnard in area 4, 3.a and 7.d is based on the NS-IBTS survey, which covers by far the largest part of the stock distribution area.

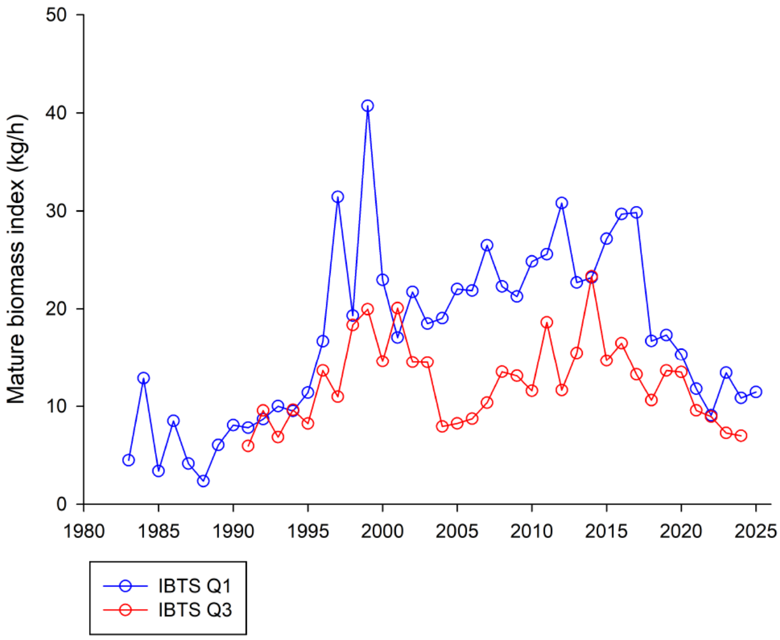

Figure 7.12. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. IBTS–Q1 and IBTS–Q3 grey gurnard mature biomass index.

7.6 MSY Proxies

7.6.1 Length Based Indicators (LBI)

Results of the length-based indicator method are sensitive to the assumed values of L_{inf} and L_{mat} (16.3 cm). During WGNSSK 2025, the estimation of L_{inf} was updated taking into account the new imported length data from commercial samplings. The estimated L_{inf} of 38.2 cm was the same as estimated one year before. How these values were estimated is described in detail in the WGNSSK 2018 report (ICES, 2018) and in the stock annex. The available length frequency distributions from InterCatch were binned into 20 mm size classes and all show a unimodal distribution (Figure 7.13). The results of the LBI method showed that, with respect to conservation, the indicators were above the reference points for L_C / L_{mat} and $L_{25\%}$ / L_{mat} for most of the data years (Figure 7.14; Table 7.2). For the $L_{max5\%}$ / L_{inf} reference point, the indicator was above the reference points since 2017. The P_{mega} was for all years below the reference of 30%. Optimum yield indicators are shown in Figure 7.15. With respect to the MSY_{proxy}, the indicator is above the reference points for the years 2017 – 2020 (Figure 7.16), but it dropped below in 2021 and 2022. In 2024, this indicator was above the defined reference.

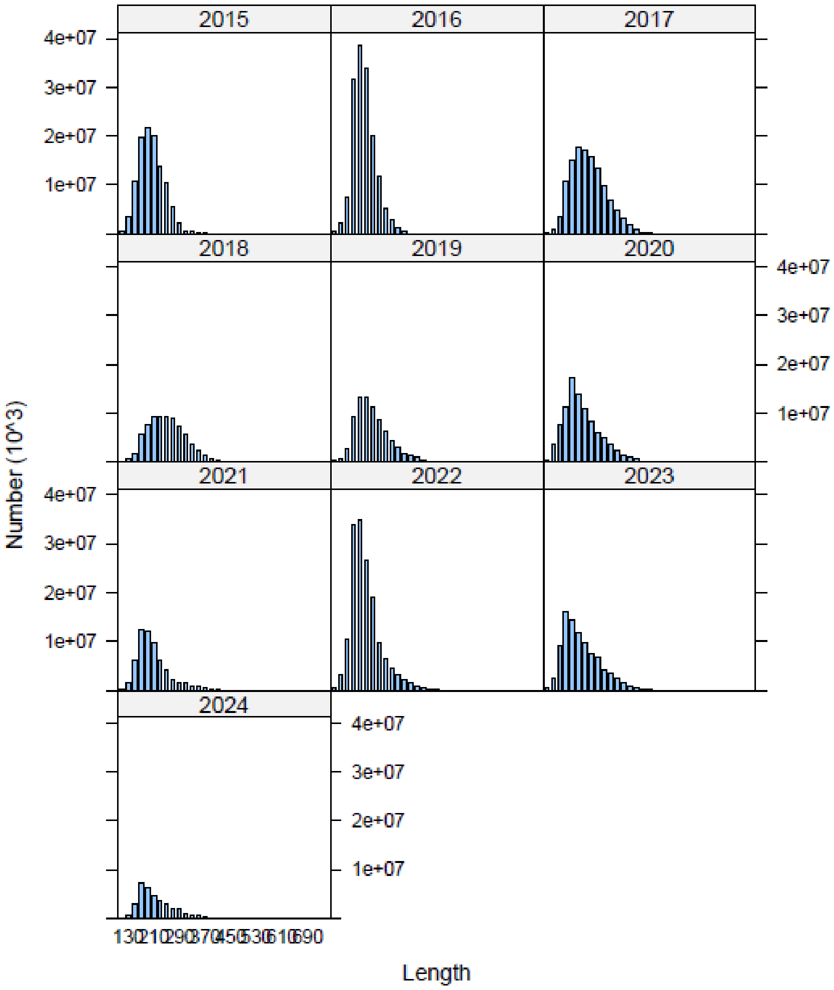

Figure 7.13 Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Obtained length frequency distributions from InterCatch binned into 20 mm size classes.

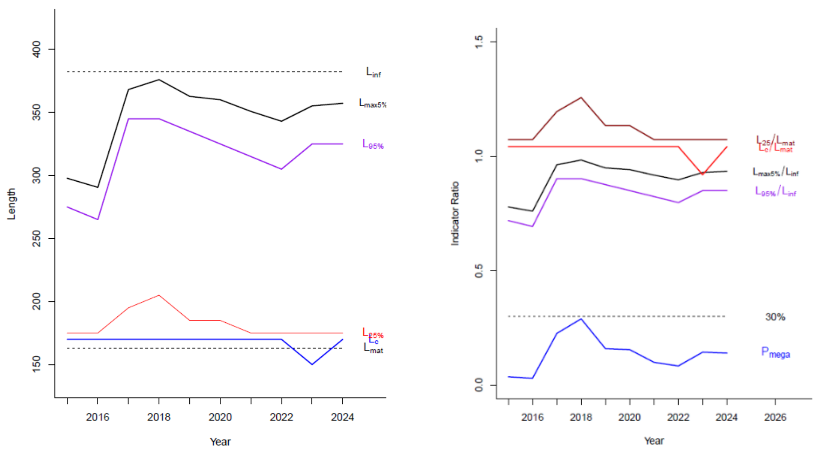

Figure 7.14 Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Conservation indicators (left panel) and indicator ratios (right panel).

Figure 7.15 Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Optimum yield indicators (left panel) and indicator ratios (right panel).

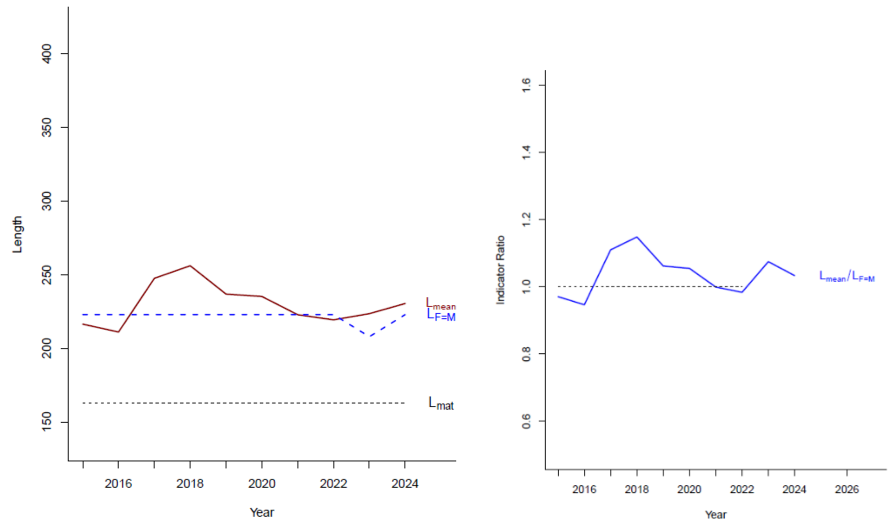

Figure 7.16 Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Maximum sustainable yield indicator (left panel) and indicator ratio (right panel).

Table 7.2 Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Length-based indicators. Green colour indicates that the observed value is above the respective reference point, red colour indicates that it is below.

	Conservation				Optimizing Yield	MSY
	L_C/L_{mat}	$L_{25\%}/L_{mat}$	$L_{max5\%}/L_{inf}$	P_{mega}	L_{mean}/L_{opt}	$L_{mean}/L_{F=M}$
Ref.:	**>1**	**>1**	**>0.8**	**>30%**	**~1(>0.9)**	**≥1**
2015	**1.04**	**1.07**	0.78	0.04	0.85	0.97
2016	**1.04**	**1.07**	0.76	0.03	0.83	0.95
2017	**1.04**	**1.20**	**0.96**	0.23	**0.97**	**1.11**
2018	**1.04**	**1.26**	**0.98**	0.29	**1.01**	**1.15**
2019	**1.04**	**1.13**	**0.95**	0.16	**0.93**	**1.06**
2020	**1.04**	**1.13**	**0.94**	0.16	**0.92**	**1.06**
2021	**1.04**	**1.07**	**0.91**	0.09	0.87	0.99
2022	**1.04**	**1.07**	**0.89**	0.08	0.86	0.98
2023	0.92	**1.07**	**0.93**	0.14	0.87	**1.07**
2024	**1.04**	**1.07**	**0.94**	0.14	**0.91**	**1.03**

7.7 Empirical *rb* rule

Not applied during WGNSSK 2025, no new catch advice due.

7.8 Data requirements

For management purposes, information should be available on catches and landings. Traditionally the quality of landings data has been poor for this species because in the past often only landings of "gurnards" were reported which is still the case for some countries today (e.g. Germany, UK England). Further, this species is highly discarded and discard data are only available for the recent years (2012–2024).

Given the high level of discarding, observations at sea under the DCF are the main source of information to better estimate total catches.

For a better understanding of this species an increase in our knowledge of biological parameters is required, especially of growth parameters with respect to the application of the *rb* rule or other empirical rules developed for data limited stocks (ICES, 2021b). In the context of ecosystem considerations, it would be useful to obtain more information on age composition of the stock and its diet composition.

From the information presented here, it can be concluded that grey gurnard is currently of very limited commercial interest.

7.9 Issues list

The available data (landings, discards, length samples) were uploaded to InterCatch for the years 2012-2024 and are used for the assessment. It should be investigated if this data series could possibly be extended to cover more years prior to 2012.

The used survey indices are well suited for this stock as the NS-IBTS covers most of the stock distribution area and shows a good catchability for this species. However, the use of a combined NS-IBTS Q1 and NS-IBTS Q3 index as well as other survey data should be investigated, e.g. applying the deltaGAM method (Berg et al., 2014).

Although catch data are incomplete for some years, it should be investigated if SPiCT (Pedersen and Berg, 2017) could serve as model to assess the grey gurnard stock, when the catch time series extends in the future.

There are some issues with the reporting of grey gurnard for some nations, e.g. Germany did not officially report grey gurnard but only a generic gurnard group in which also other gurnard species are included. The latter is also often the case in logbook data. This was corrected when uploading data to InterCatch, by using the ratio of gurnard species in Dutch and Belgian official landings (20% of gurnards landed were grey gurnards), assuming a similar landing pattern for the German fleet. For UK data a ratio from survey data was often used in the past to correct for the proportion of other gurnard species. However, these methods will introduce additional uncertainties in the final catch estimates.

For some fleets zero landings are reported, but without any information about possible discards (e.g., Swedish OTB_DEF_>=120_0_0_0_all). For these cases it is not possible to raise any discards in InterCatch, although high discards may occur in these fleets. It is not known how this affects the estimation of the total catch within InterCatch.

Biological data are not collected on a routine basis for grey gurnard on the NS-IBTS. However, from time to time new data are available via DATRAS and the availability of these data, and possibly also other data sources, should be checked and compiled during a benchmark process.

7.10 References

Berg, C., Nielsen, A., Christensen, K., 2014. Evaluation of alternative age-based methods for estimating relative abundance from survey data in relation to assessment models. Fisheries Research 151: 91-99.

Daan, N., Bromley, P. J., Hislop, J. R. G., and Nielsen, N. A., 1990. Ecology of North Sea Fish. Netherlands Journal of Sea Research 26(2–4): 343–386.

de Gee, A., and Kikkert, A.H., 1993. Analysis of grey gurnard (Eutrigla gurnardus) samples collected during the 1991 International Stomach Sampling Project. ICES CM/G:14. 25 pp.

Floeter, J., Temming, A., 2005. Analysis of prey size preference of North Sea whiting, saithe, and grey gurnard. ICES Journal of Marine Science 62: 897–907.

Hertling, H. 1924. Über den grauen und den roten Knurrhahn (Trigla gurnardus L. und Trigla hirundo Bloch). Wissenschaftliche Meeresuntersuchungen Helgoland 15(2), Abhandlung 13: 1–53.

ICES 2011. Report of the Joint ICES STECF Workshop on Management Plan Evaluations for Roundfish Stocks (WKROUNDMP/EWG 11–01), 28 February–4 March 2011, ICES Headquarters, Copenhagen: 67 pp.

ICES 2012. Report of the Working Group on Assessment of New MoU Species (WGNEW). ICES CM 2012/ACOM:20.

ICES 2014. Report of the Working Group on Assessment of New MoU Species (WGNEW), 24–28 March 2014, ICES Headquarters, Denmark. ICES CM 2014/ACOM:21.

ICES 2015. Report of the Working Group on the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK), 24 April–5 May 2015, ICES Headquarters, Copenhagen. ICES CM 2015/ACOM:13.

ICES 2017. Interim Report of the Working Group on Multispecies Assessment Methods (WGSAM), 16-20 October 2017, San Sebastian, Spain. ICES CM 2017/SSGEPI:20.

ICES 2018. Report of the Working Group on the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK), 24 April-03 May 2018, Oostende, Belgium.

ICES 2020. Working Group on the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK). ICES Scientific Reports. 2:61. 1140 pp. http://doi.org/10.17895/ices.pub.6092

ICES 2021a. International Bottom Trawl Survey Working Group (IBTSWG). ICES Scientific Reports. 3:69. 201 pp. https://doi.org/10.17895/ices.pub.8219

ICES 2021b. Tenth Workshop on the Development of Quantitative Assessment Methodologies based on LIFE-history traits, exploitation characteristics, and other relevant parameters for data-limited stocks (WKLIFE X). ICES Scientific Reports. 2:98. 72 pp. http://doi.org/10.17895/ices.pub.5985

ICES 2022. ICES technical guidance for harvest control rules and stock assessments for stocks in categories 2 and 3. In Report of ICES Advisory Committee, 2022. ICES Advice 2022, Section 16.4.11. https://doi.org/10.17895/ices.advice.19801564

Kempf, A., Stelzenmüller, V., Akimova, A., Floeter, J., 2013. Spatial assessment of predator-prey relationship in the North Sea: the influence of abiotic habitat properties on the spatial overlap between 0–group cod and grey gurnard. Fisheries Oceanography 22(3):174–192.

Pedersen, M. W., Berg C. W., 2017. A stochastic surplus production model in continuous time. Fish and Fisheries, 18: 226-243. DOI: 10.1111/faf.12174.

Sahrhage, D., 1964. Über die Verbreitung der Fischarten in der Nordsee. I. Juni–Juli 1959 und Juli 1960. Berichte der Deutschen Wissenschaftlichen Kommission für Meeresforschung 17(3): 165–278.

Ulleweit, J., Stransky, C., Panten, K., 2010. Discards and discarding practices in German fisheries in the North Sea and Northeast Atlantic during 2002–2008. Journal of Applied Ichthyology 26(1): 54–66.

Uhlmann, S.S., Coers, A., van Helmond, A.T.M., Nijman, R.R., Bol, R.A., van Reijden, K., 2013. Discard sampling of Dutch bottom-trawl and seine fisheries in 2012. CVO report: 13.015. 76pp.

van Heessen and Daan, 1996. Long-term trends in ten non-target North Sea fish species. ICES Journal of Marine Science, 53: 1063–1078.

7.11 Catch and index tables

Table 7.3. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Official grey gurnard landings in Division 3.a (tonnes).

Year	BE	DK	NL	NO	SE	Total
1980	0	0	0	0	36	36

Year	BE	DK	NL	NO	SE	Total
1981	0	0	0	0	46	46
1982	0	86	0	0	43	129
1983	0	29	0	0	7	36
1984	0	62	0	0	6	68
1985	0	3	0	0	9	12
1986	0	6	0	0	10	16
1987	1	13	0	0	6	20
1988	0	59	0	0	2	61
1989	0	19	0	0	4	23
1990	0	34	0	0	3	37
1991	0	25	0	0	5	30
1992	0	22	0	0	10	32
1993	0	18	0	0	9	27
1994	0	12	0	0	12	24
1995	0	10	0	0	5	15
1996	0	18	0	0	3	21
1997	0	13	0	0	5	18
1998	0	27	0	0	8	35
1999	0	23	0	0	5	28
2000	0	32	0	0	5	37
2001	0	30	0	0	3	33
2002	0	18	0	0	1	19
2003	0	32	0	0	1	33
2004	0	24	2	0	2	28
2005	0	21	4	0	1	26
2006	0	19	0	0	2	21
2007	0	21	1	0	3	25
2008	0	24	0	0	5	29
2009	0	15	0	0	3	18

Year	BE	DK	NL	NO	SE	Total
2010	0	10	1	0	2	13
2011	0	5	0	0	1	6
2012	0	5	0	0	1	6
2013	0	5	0	0	1	6
2014	0	3	0	0	1	4
2015	0	10	1	1	2	14
2016	0	13	1	0	2	16
2017	0	257	6	4	3	270
2018	0	24	11	0	3	38
2019	0	7	10	0	2	19
2020	0	4	15	0	1	20
2021	0	3	13	0	1	17
2022	0	18	2	0	1	21
2023*	0	3	0	0	1	4
2024*	0	6	2	0	1	9

*** preliminary data**

Table 7.4. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Official grey gurnard landings in Subarea 4 (tonnes).

Year	BE	DK	FR	NL	NO	SE	UK	FO	DE	Total
1980	0	0	43	0	0	0	0			43
1981	0	0	0	0	0	0	0			0
1982	0	0	100	0	0	0	0			100
1983	0	0	64	0	0	0	0			64
1984	0	0	71	0	0	0	0			71
1985	88	0	85	0	0	0	0			173
1986	0	27	66	0	0	0	0			93
1987	63	44205	56	0	0	0	0			44324
1988	72	36887	43	0	0	0	22			37024
1989	73	26230	45	0	0	0	0			26348

Year	BE	DK	FR	NL	NO	SE	UK	FO	DE	Total
1990	85	22041	42	0	0	0	0			22168
1991	70	14514	28	0	0	0	0			14612
1992	98	8113	21	0	0	0	10			8242
1993	106	822	27	0	0	0	24			979
1994	63	87	21	0	0	0	22			193
1995	43	63	26	0	0	0	21			153
1996	108	52	18	0	0	0	54			232
1997	49	23	22	0	0	0	57			151
1998	33	29	13	0	0	0	0			75
1999	35	63	0	0	0	127	0			225
2000	28	63	5	452	0	0	0			548
2001	22	258	20	277	0	1	33			611
2002	23	45	10	285	0	1	29			393
2003	16	60	5	307	0	6	26			420
2004	21	59	6	264	0	3	23			376
2005	16	52	5	213	0	8	22			316
2006	10	46	2	133	2	0	7			200
2007	11	16	3	155	5	0	14			204
2008	8	24	2	104	5	3	12			158
2009	15	6	2	154	1	1	22			201
2010	14	8	10	218	2	0	14			266
2011	26	6	7	263	1	0	31			334
2012	49	3	4	467	2	0	77			602
2013	30	5	2	268	33	1	131			470
2014	35	4	3	252	56	0	128			478
2015	22	1220	2	229	172	5	354			2004
2016	31	1151	6	232	83	6	297			1806
2017	24	2068	4	329	172	8	320			2925
2018	27	497	14	364	149	16	463			1530

Year	BE	DK	FR	NL	NO	SE	UK	FO	DE	Total
2019	26	301	3	425	203	51	567			1576
2020	25	505	2	440	276	21	471	1	46	1787
2021	10	151	7	463	125	6	304			1066
2022	2	139	1	250	97	2	260	1	1	753
2023*	2	221	3	176	36	6	413		2	859
2024*	4	168	1	110	31	15	364		3	696

* preliminary data

Table 7.5. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Official grey gurnard landings in Division 7.d (tonnes).

Year	BE	FR	NL	UK	Total
1980	0	950	0	0	950
1981	0	0	0	0	0
1982	0	380	0	0	380
1983	0	489	0	0	489
1984	0	126	0	0	126
1985	14	102	0	0	116
1986	0	217	0	0	217
1987	12	66	0	0	78
1988	14	346	0	0	360
1989	9	90	0	0	99
1990	6	92	0	0	98
1991	5	94	0	0	99
1992	6	85	0	0	91
1993	7	47	0	0	54
1994	4	33	0	0	37
1995	7	36	0	0	43
1996	4	44	0	0	48
1997	3	81	0	0	84
1998	1	34	0	0	35
1999	1	0	0	0	1

Year	BE	FR	NL	UK	Total
2000	9	67	0	0	76
2001	6	40	0	0	46
2002	32	54	1	0	87
2003	18	42	12	0	72
2004	14	3	31	0	48
2005	13	2	21	0	36
2006	8	2	22	14	46
2007	3	1	9	36	49
2008	1	3	16	66	86
2009	1	1	3	61	66
2010	6	2	39	64	111
2011	11	5	53	33	102
2012	11	5	11	23	50
2013	23	4	11	14	52
2014	7	5	4	2	18
2015	2	6	2	0	10
2016	1	6	2	0	9
2017	1	8	4	12	25
2018	17	6	4	11	38
2019	1	7	3	8	19
2020	1	2	1	1	5
2021	1	3	2	1	7
2022	1	1	1	1	4
2023*	2	3	1	0	6
2024*	2	3	1	0	6

*** preliminary data**

Table 7.6. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Mature biomass indices (kg/hour) from IBTS–Q1 and IBTS–Q3.

Year	IBTS–Q1	IBTS–Q3
1983	4.48	

Year	IBTS–Q1	IBTS–Q3
1984	12.85	
1985	3.38	
1986	8.49	
1987	4.15	
1988	2.35	
1989	6.03	
1990	8.07	
1991	7.80	5.93
1992	8.67	9.55
1993	10.01	6.84
1994	9.51	9.62
1995	11.38	8.22
1996	16.68	13.63
1997	31.44	10.96
1998	19.31	18.35
1999	40.72	19.96
2000	22.97	14.59
2001	17.06	20.08
2002	21.72	14.53
2003	18.49	14.52
2004	19.05	7.93
2005	22.02	8.23
2006	21.87	8.71
2007	26.49	10.35
2008	22.29	13.52
2009	21.26	13.10
2010	24.85	11.56
2011	25.59	18.63
2012	30.81	11.64
2013	22.70	15.47

Year	IBTS–Q1	IBTS–Q3
2014	23.20	23.33
2015	27.16	14.68
2016	29.69	16.49
2017	29.84	13.24
2018	16.71	10.61
2019	17.32	13.64
2020	15.32	13.49
2021	11.78	9.56
2022	9.07	8.93
2023	13.41	7.27
2024	10.84	6.98
2025	11.44	

Table 7.7. Grey gurnard in Subarea 4 and divisions 7.d and 3.a. Official landings, BMS landings, ICES landings (incl. IBC), discards (incl. BMS), and catches in tonnes. *Preliminary

Year	Official landings	Official BMS landings	ICES Landings	ICES catches	ICES discards	Discard rate
1983	589					
1984	265					
1985	301					
1986	326					
1987	44422					
1988	37445					
1989	26470					
1990	22303					
1991	14741					
1992	8365					
1993	1060					
1994	254					
1995	211					
1996	301					

Year	Official landings	Official BMS landings	ICES Landings	ICES catches	ICES discards	Discard rate
1997	253					
1998	145					
1999	254					
2000	661					
2001	690					
2002	499					
2003	525					
2004	452					
2005	378					
2006	267					
2007	278					
2008	273					
2009	285					
2010	390					
2011	442					
2012	658		689	8345	7656	0.92
2013	528		1180	10230	9050	0.88
2014	500		1892	8596	6704	0.78
2015	2028		2141	8452	6311	0.75
2016	1831		2156	12058	9902	0.82
2017	3220		3451	16940	13489	0.80
2018	1607		1137	11418	10281	0.90
2019	1614	13	1709	9295	7586	0.82
2020	1756	6	1966	10221	8255	0.81
2021	1090	0	1099	5549	4450	0.80
2022	778	0	780	9886	9106	0.92
2023	869*	0	879	9559	8680	0.91
2024	711*	0	707	3261	2554	0.78

Contents